• Introduction

- This project involves analyzing a real-world dataset of delivery orders to understand the root causes of delays.
The goal is to provide data-driven insights that can help the company enhance its service quality, improve customer satisfaction, and increase profitability.

- To achieve this, we performed the following steps:

• Data Cleaning
• Exploratory Data Analysis (Visualization)
• Decision Tree Modeling
• K-means clustering
• Insights
• Shiny App Development

- The main goals are to:

• Identify factors causing delays
• Build a predictive model
• Cluster deliveries.

- Dataset Description:

- The dataset used in this project contains **912 delivery records** collected from a food delivery service.

- It includes both numerical and categorical variables that describe the delivery process, environmental conditions, and courier characteristics.

- The main purpose of this dataset is to analyze factors that influence delivery time and late delivery behavior.

Order_ID Unique identifier for each delivery order.	Distance_km Distance between the restaurant and the delivery address (in kilometers).	Weather Weather conditions during the delivery	Traffic_Level Traffic conditions during the delivery	Time_of_Day Time of the day when the delivery took place	Vehicle Type of v the delive
1 — 1000	0.59 — 20 (14.17 - 16.11, Count: 89)	Clear 47% Rainy 20% Other (326) 33%	Medium 39% Low 38% Other (227) 23%	Morning 31% Evening 29% Other (399) 40%	Bike Scooter Other (19
522	7.93	Windy	Low	Afternoon	Scooter
738	16.42	Clear	Medium	Evening	Bike
741	9.52	Foggy	Low	Night	Scooter
661	7.44	Rainy	Medium	Afternoon	Scooter
412	19.03	Clear	Low	Morning	Bike
679	19.4	Clear	Low	Evening	Scooter
627	9.52	Clear	Low		Bike
514	17.39	Clear	Medium	Evening	Scooter
860	1.78	Snowy	Low	Evening	Car
137	10.62	Foggy	Low	Evening	Scooter
812	16.86	Snowy	Medium	Afternoon	Car
77	15.54	Clear	Low	Night	Bike

• Data Cleaning:

- Before performing any analysis or modeling, the dataset was examined and cleaned to ensure accuracy, consistency, and suitability for machine learning tasks. The data cleaning process included several key steps, described as follows:

• Removal of Irrelevant Rows
• Handling Data Types
  → numeric values is replaced by mean
  → string value is replaced to NA then omitted
• Checking for Missing Values
• Outlier Inspection
• Adding new features to the data base to enhance the analysis like (speed, late_order)
• Ensuring Overall Consistency

	X	Order_ID	Distance_km	Weather	Traffic_Level	Time_of_Day	Vehicle_Type	Preparation_Time_min	Courier_Experience_yrs	Delivery_Time_min	Total_time	late_order	Speed
1	1	522	7.93	Windy	Low	Afternoon	Scooter	12	1.000000	43	55	Not late	11.065116
2	2	738	16.42	Clear	Medium	Evening	Bike	20	2.000000	84	104	Late	11.728571
3	3	741	9.52	Foggy	Low	Night	Scooter	28	1.000000	59	87	Late	9.681356
4	4	661	7.44	Rainy	Medium	Afternoon	Scooter	5	1.000000	37	42	Not late	12.064865
5	5	412	19.03	Clear	Low	Morning	Bike	16	5.000000	68	84	Late	16.791176
6	6	679	19.40	Clear	Low	Evening	Scooter	8	9.000000	57	65	Late	20.421053
7	8	514	17.39	Clear	Medium	Evening	Scooter	5	6.000000	46	51	Not late	22.682609
8	9	860	1.78	Snowy	Low	Evening	Car	20	6.000000	35	55	Not late	3.051429
9	10	137	10.62	Foggy	Low	Evening	Scooter	29	1.000000	73	102	Late	8.728767
10	11	812	16.86	Snowy	Medium	Afternoon	Car	13	4.000000	88	101	Late	11.495455
11	12	77	15.54	Clear	Low	Night	Bike	29	1.000000	76	105	Late	12.268421
12	13	637	10.89	Clear	High	Night	Bike	12	5.000000	53	65	Not late	12.328302
13	14	974	4.69	Rainy	Medium	Evening	Scooter	12	7.000000	36	48	Not late	7.816667
14	15	939	2.80	Clear	High	Morning	Scooter	10	4.579381	33	43	Not late	5.090909
15	16	900	2.17	Snowy	Low	Evening	Car	15	3.000000	35	50	Not late	3.720000
16	17	281	17.86	Rainy	Low	Afternoon	Bike	5	1.000000	50	55	Not late	21.432000
17	18	884	2.53	Snowy	Low	Morning	Bike	6	8.000000	24	30	Not late	6.325000
18	19	762	6.58	Foggy	Low	Evening	Car	6	7.000000	27	33	Not late	14.622222
19	20	320	5.38	Clear	Medium	Evening	Scooter	26	1.000000	47	73	Not late	6.868085
20	21	550	15.56	Rainy	Low	Afternoon	Scooter	28	0.000000	72	100	Late	12.966667
21	22	175	13.08	Rainy	Low	Morning	Car	16	4.000000	58	74	Late	13.531034
22	23	372	0.86	Clear	High	Afternoon	Bike	26	5.000000	37	63	Not late	1.394595
23	24	528	5.83	Clear	High	Evening	Car	27	9.000000	56	83	Not late	6.246429
24	25	211	11.20	Clear	Medium	Morning	Bike	23	4.579381	73	96	Late	9.205479

Showing 1 to 25 of 912 entries, 13 total columns

• Exploratory Data Analysis:

- To understand the behavior of the dataset before applying machine learning models, several visualizations were created. These plots helped identify patterns, trends, and possible factors that may influence late deliveries. the main visualizations and their observations are summarized below:

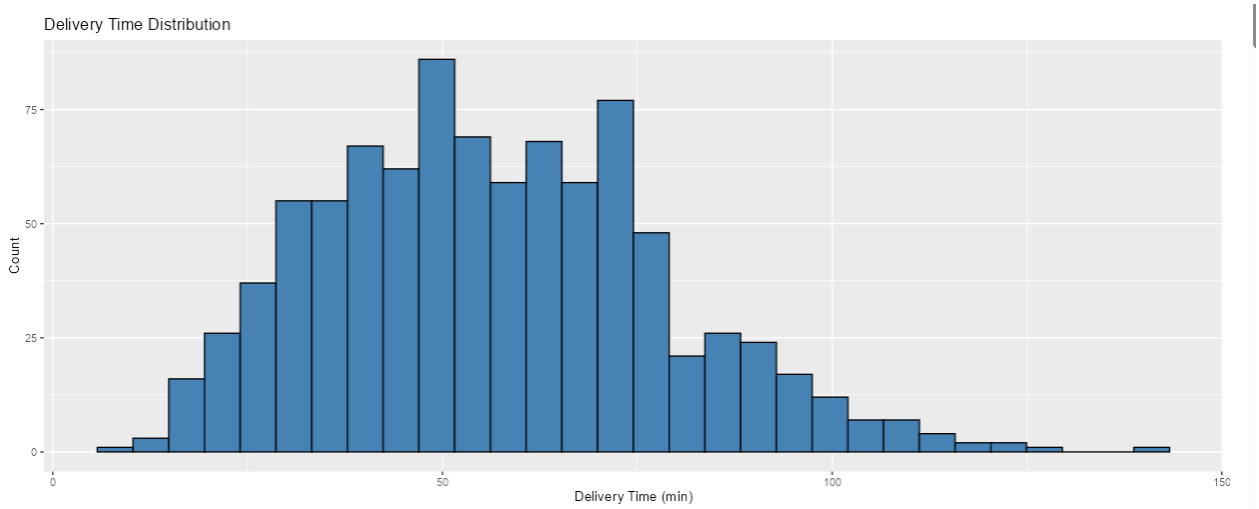
• **Delivery Time Distribution:**
- A histogram was used to show how delivery times are spread across all orders.
• **Distance Distribution:**
- A histogram of [Distance_km] shows how far orders were delivered.
• **Preparation Time Distribution:**
- A histogram of preparation time was plotted.
• **Delivery Time by Vehicle Type:**
- A boxplot was created to compare delivery times across different vehicle types.
• **Delivery Time by Traffic Level:**
- A boxplot was used to study how traffic affects delivery time
• **Delivery Time by Weather Condition:**
- A boxplot comparing delivery times under different weather conditions was created.

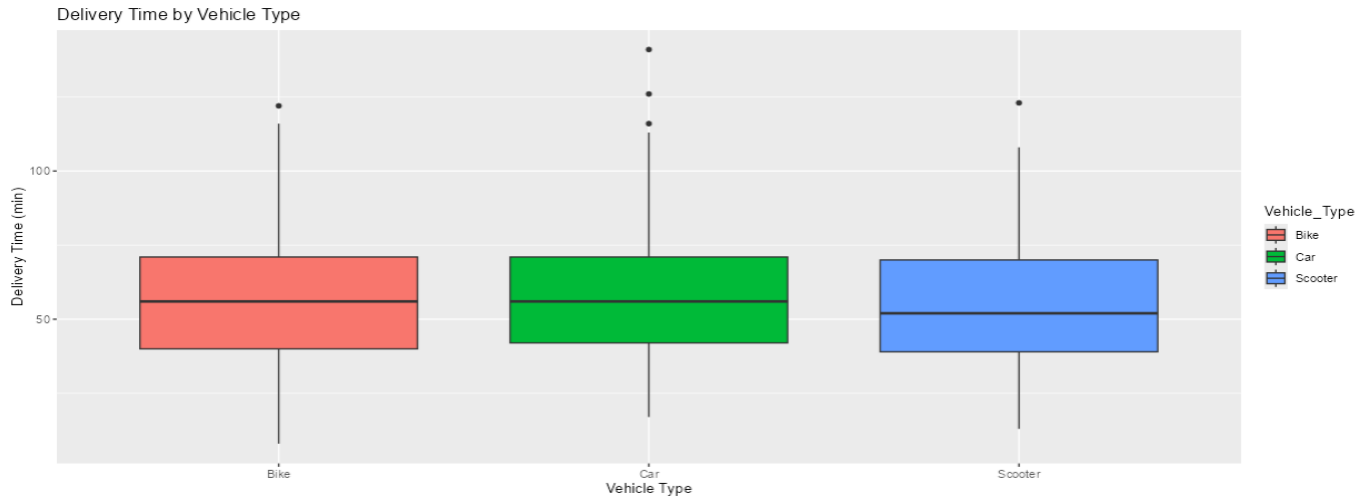
- **Distance vs Delivery Time:**

- A scatter plot visualizing the relationship between distance and delivery time.

- **Preparation Time vs Delivery Time:**

- A scatter plot visualizing preparation time against delivery time.





• Decision Tree Model:

- The Decision Tree model was used to predict whether a delivery would be late based on operational and environmental factors. this model was selected because it provides a clear and interpretable structure that highlights the most important variables contributing to late deliveries.

• Model Setup:

• The dataset was split into:

- 70% training data • 30% testing data

• The model predicts the variable:

- late_order (Late/Not Late)

• The main features used in the model were:

- Distance_km - Preparation_Time_min - Weather
- Traffic_Level - Time_of_Day - Vehicle_Type
- Courier_Experience_yrs

These features describe delivery conditions, courier experience, and restaurant processing time.

• Model Results:

- After training the model, predictions were made on the test data.

- A confusion matrix was generated

```
Confusion Matrix:
          Predicted
Actual     Late Not late
  Late      107       23
  Not late   14      127
```

- The model achieved an accuracy of **86.34686%**

```
Accuracy:
[1] 0.8634686
```

This accuracy shows how well the model can classify orders as late or on time.

• Interpretation of the Tree:

- Distance_km:

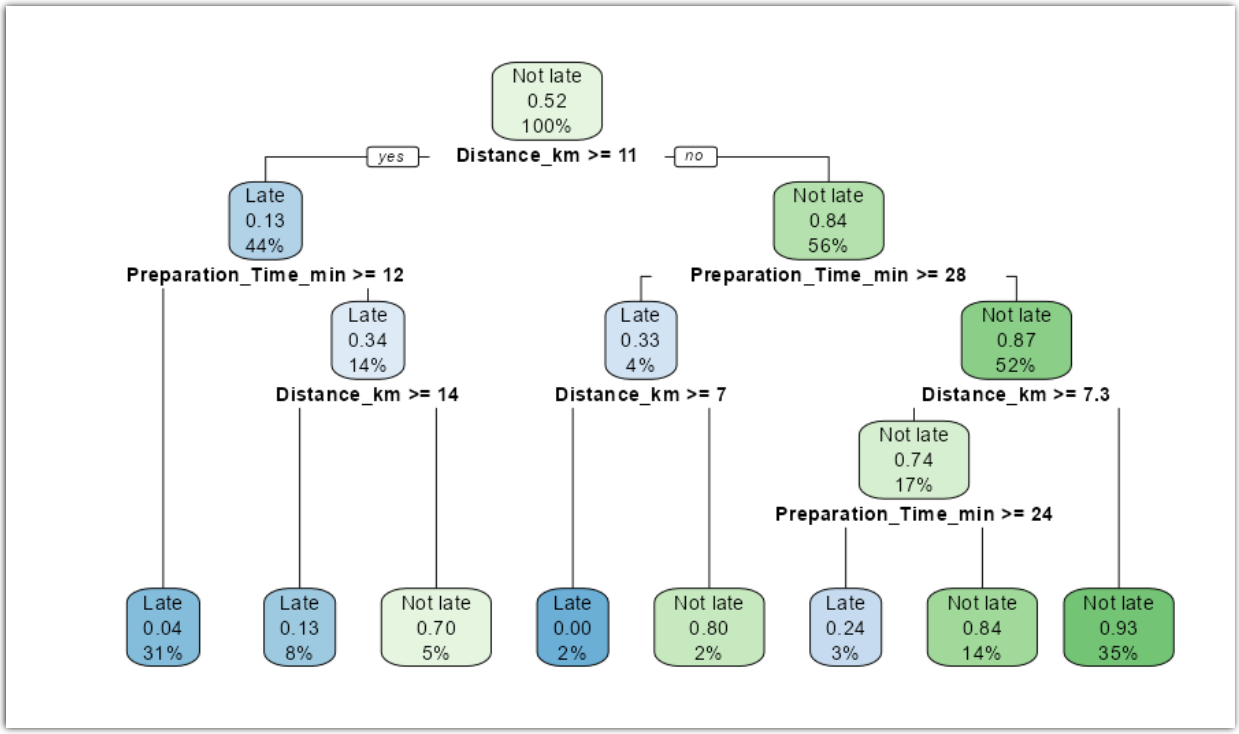
The decision tree showed that this was the most important factor. Longer distances strongly increase the chance of a late delivery.

- Preparation_Time_min:

also this one was the second most important factor. Orders that take longer to prepare often become late.

- Traffic and Weather:

also affect delivery time but have a smaller impact compared to distance and preparation time.



- Example Rule:

- If the distance is high
  AND preparation time is long → The order is likely to be late

• K-means Clustering:

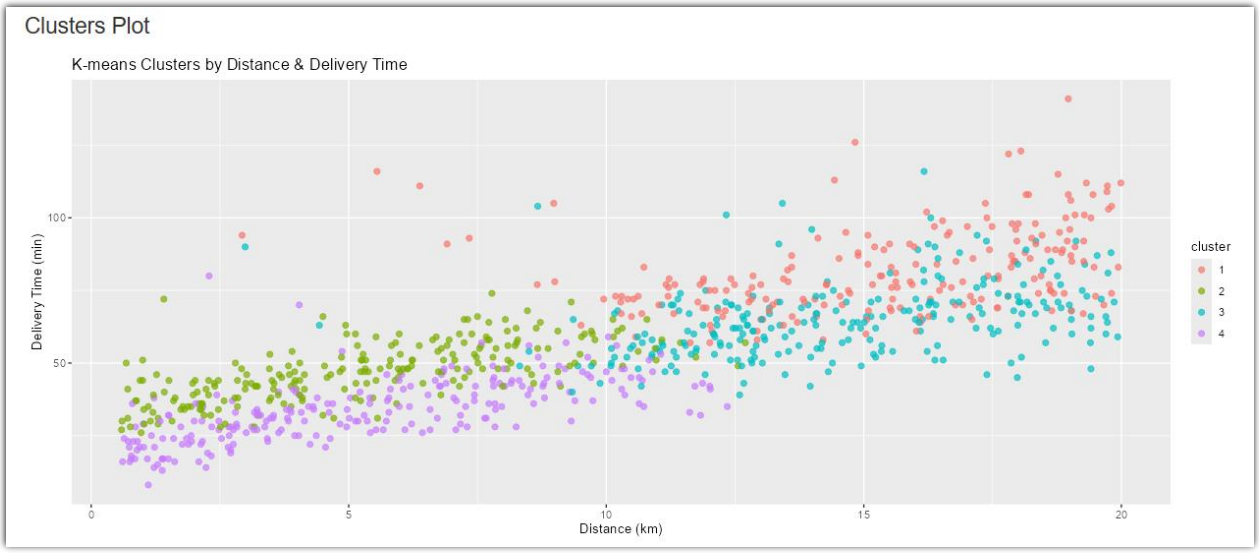
- K-means clustering was used to group delivery orders into meaningful categories based on their numerical characteristics. This helps us understand different delivery patterns and identify types of orders that are more likely to be fast or delayed.

- We selected four numerical features that best describe the delivery process:

• Distance_km
• Delivery_Time_min
• Preparation_Time_min
• Courier_Experience_yrs

Before running K-means, all features were standardized using the scale() function.

- The Shiny App allows the user to interactively select the number of clusters (K). For analysis, we explored 3 to 4 clusters which aligns with the project requirements.



A scatter plot of **Distance_km vs. Delivery_Time_min** was used to visualize the clusters. Each point is colored according to its assigned cluster, making the grouping pattern clear.

• The visualization shows:

- Short-distance orders cluster together at lower delivery times.
- Long-distance orders form separate clusters with higher delivery times.
- Preparation time also influences cluster separation.
- Clusters are well spaced, indicating good segmentation.

Cluster Summary

cluster	avg_distance	avg_delivery_time	avg_prep_time	avg_experience	n_orders
1	15.27	81.79	23.71	4.70	208
2	5.31	47.08	23.01	3.85	228
3	14.52	64.89	11.06	4.16	246
4	5.32	33.80	11.30	5.88	230

- K-means successfully grouped the delivery orders into distinct clusters. type of delivery behavior.
- Based on the cluster summary statistics, the clusters can be interpreted as follows:

• Cluster 1:

Longest distances - High preparation time - High preparation time → Represents **slow and high-risk deliveries**

• Cluster 2:

Short distances - Medium delivery time - High preparation time → Represents **average-speed deliveries affected mainly by preparation delays**

• Cluster 3:

Long distance - Moderate delivery time - Lower preparation time → Represents **efficient long-distance deliveries**

• Cluster 4:

Short distance - Shortest delivery time - Faster preparation time → Represents **fast and low-risk deliveries**

• Key Analysis Questions:

1) What are the primary factors that cause late deliveries ?

• Distance + Preparation Time + Traffic + Weather.

2) Can delivery orders be grouped into meaningful clusters? What do these clusters represent?

• Yes. K-means clustering created clear and meaningful segments of delivery behavior:
Cluster 1: Long-distance, slow deliveries (high-risk)
Cluster 2: Short-distance, average deliveries
Cluster 3: Long-distance but efficient deliveries due to faster preparation
Cluster 4: Short-distance, fast deliveries (low-risk)

• These clusters help identify different types of orders and can guide resource allocation.

3) Can we predict a late delivery before the order is dispatched?

• The Decision Tree model was able to predict late deliveries with an accuracy of **86.34686%**

The model's rules show:

- Deliveries with **high distance** and **long preparation time** are most likely to be late.
- The tree structure makes the prediction easy to interpret and actionable.

4) What specific actions can the company take to reduce delays?

- Based on the insights, the company can take several steps:

- **Reduce preparation time** by improving kitchen workflow.
- **Assign faster vehicles** (e.g., scooters) to long-distance orders.
- **Increase staffing** during high-traffic or night periods.
- **Monitor weather conditions** and adjust delivery estimates.
- **Use clustering** to identify high-risk orders early and prioritize them.

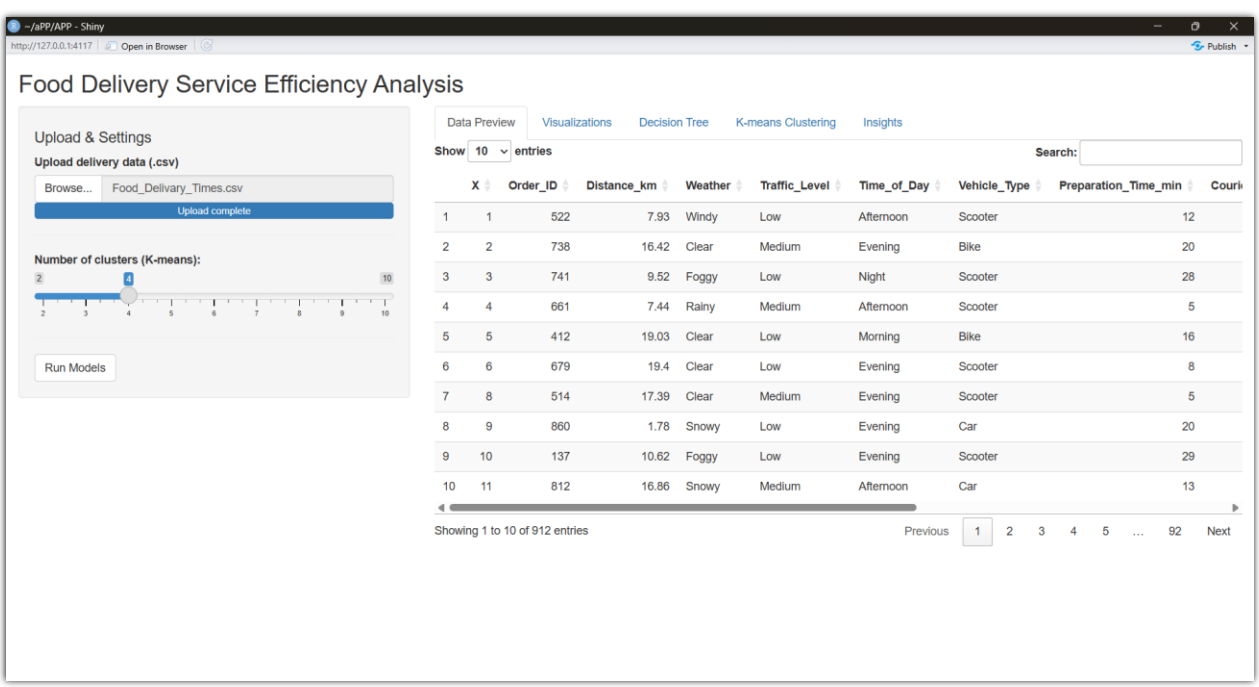
• Shiny App (GUI):

A Shiny web application was developed to make the analysis interactive and easier to explore.

The app allows users to upload the dataset, view the data, generate visualizations, run models, and interpret results in a simple and user-friendly way.

• Main Features of the Application:

- File Upload
- Data Preview
- Visualizations
- Decision Tree Model
- K-means Clustering
- Insights Tab



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4	4	661	7.44	Rainy	Medium	Afternoon	Scooter	5	
5	5	412	19.03	Clear	Low	Morning	Bike	16	
6	6	679	19.4	Clear	Low	Evening	Scooter	8	
7	8	514	17.39	Clear	Medium	Evening	Scooter	5	
8	9	860	1.78	Snowy	Low	Evening	Car	20	
9	10	137	10.62	Foggy	Low	Evening	Scooter	29	
10	11	812	16.86	Snowy	Medium	Afternoon	Car	13	

• Purpose of the GUI:

- Explore the dataset freely.
- Visualize important relationships.
- Run the decision tree and clustering models instantly.
- Understand the analysis results without writing code.
- Support better decision-making for improving delivery efficiency.

• Insights:

1) Distance_km and Preparation_Time_min are strong predictors of late deliveries.

2) Long distances combined with long preparation times lead to a high probability of late orders.

3) K-means clustering shows different groups of orders (short & fast, long & risky, etc.).

4) The company can reduce delays by improving preparation time and assigning suitable vehicles for long distances.